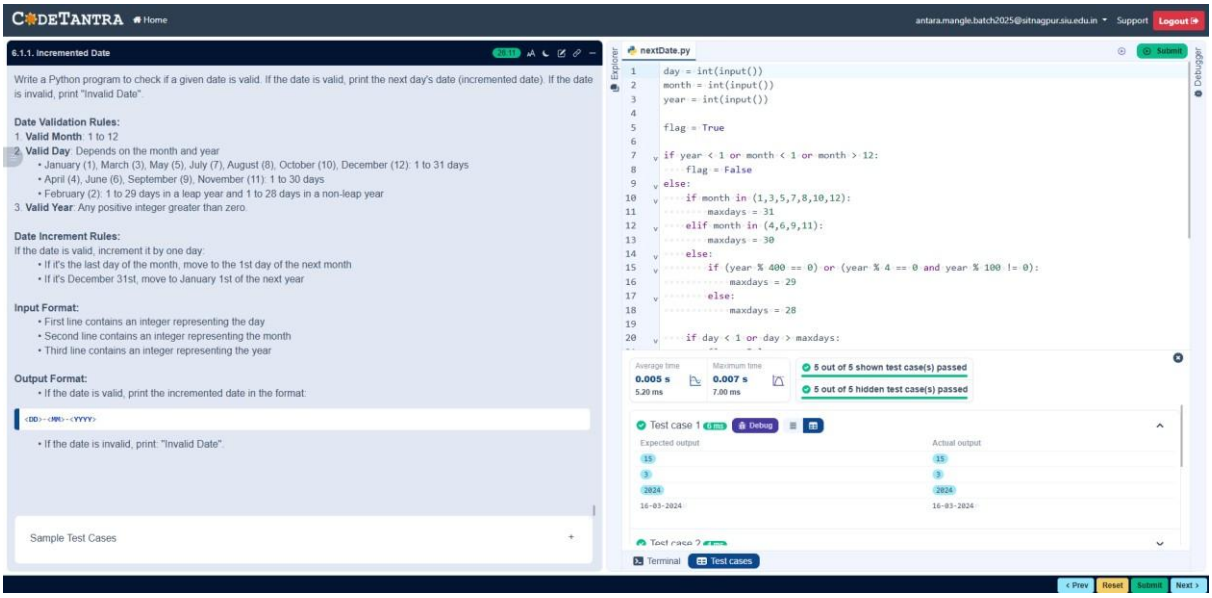
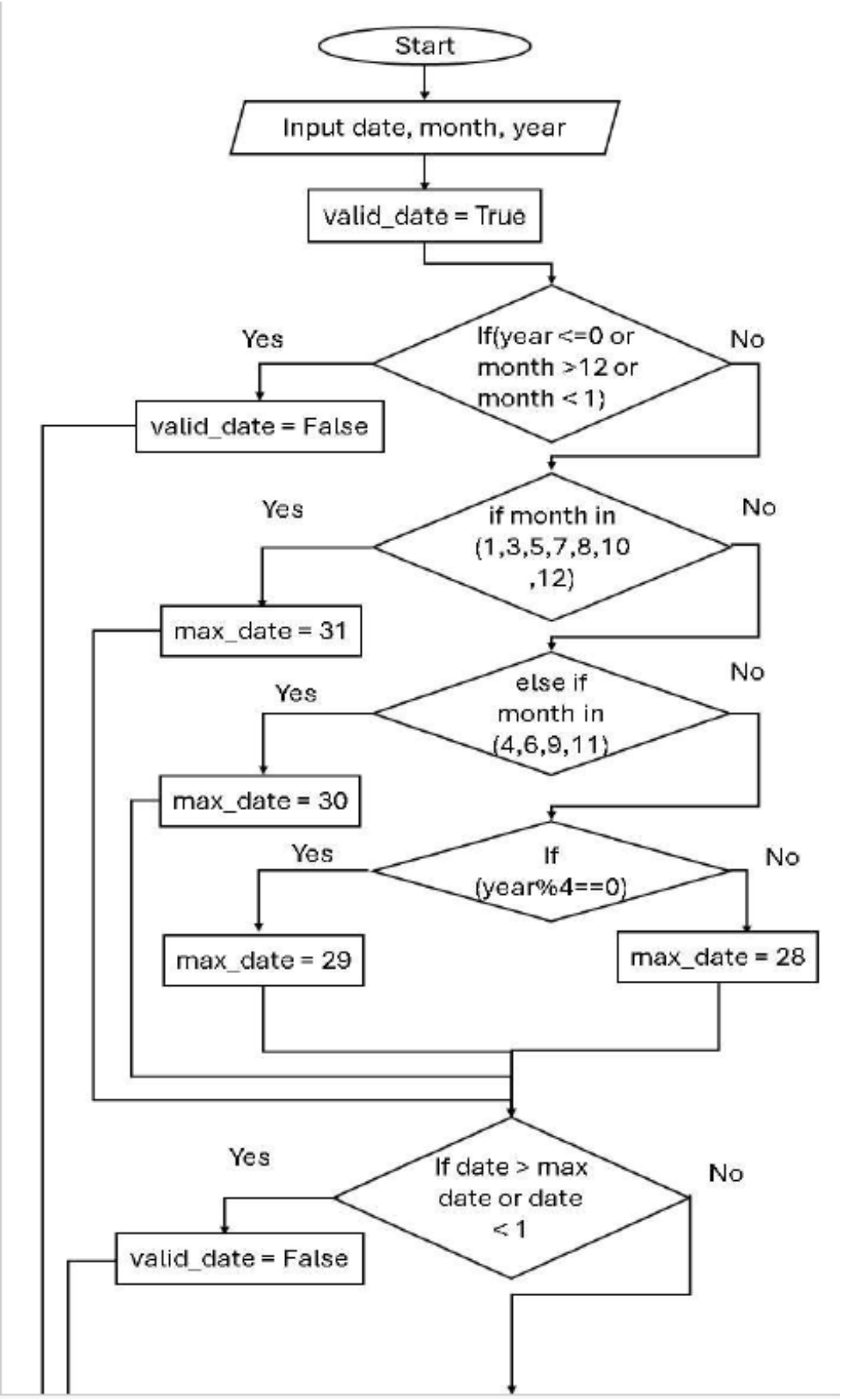
6.1

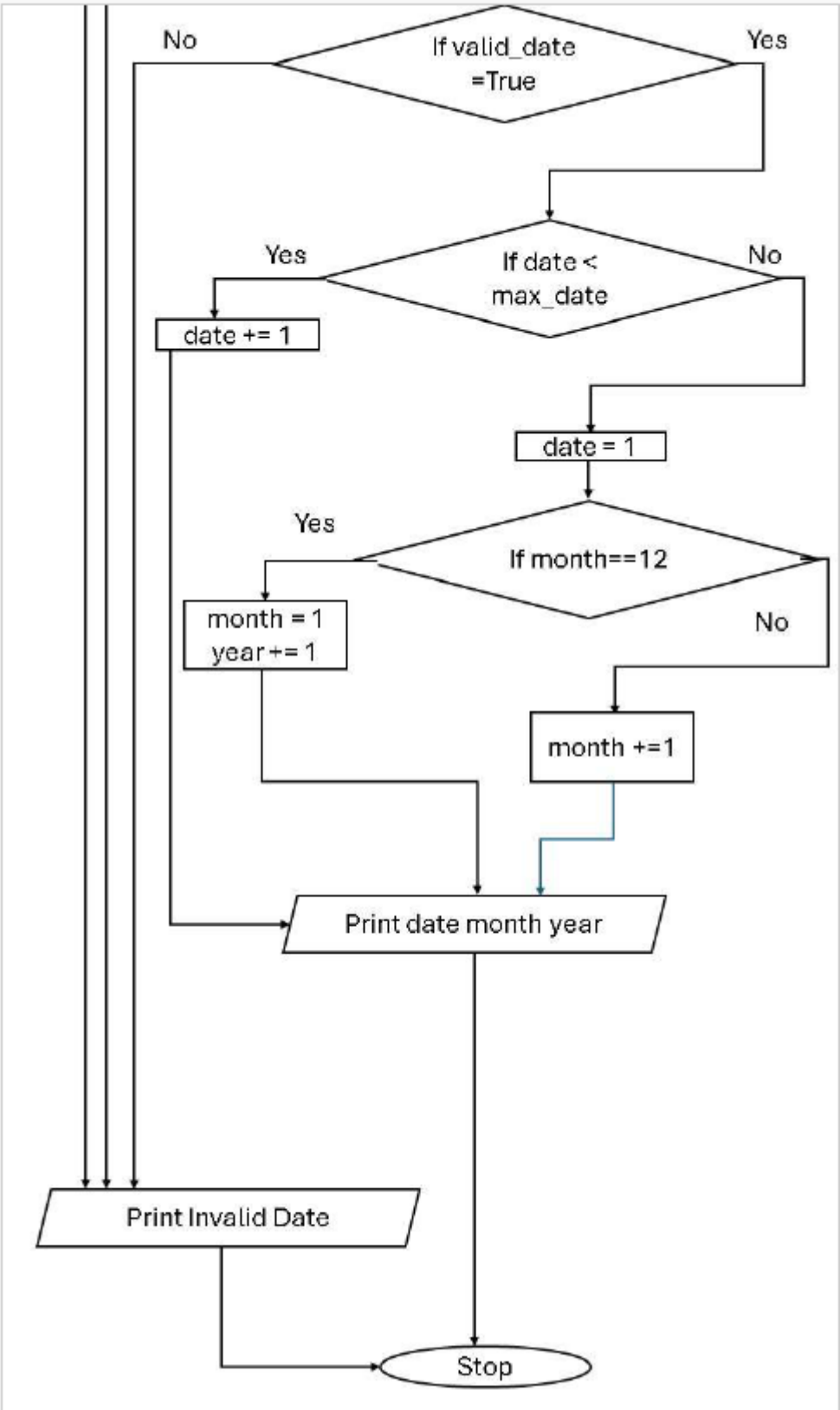


Algorithm

1. Start.
2. Read day, month, and year from the user.
3. If year ≤ 0 → Print "Invalid Date" → Stop
4. If month < 1 OR month > 12 → Print "Invalid Date" → Stop
5. If (year % 400 == 0) → Set leap = True
6. Else if (year % 100 == 0) → Set leap = False
7. Else if (year % 4 == 0) → Set leap = True
8. Else → Set leap = False
9. If month is 1, 3, 5, 7, 8, 10, or 12 → Set max_day = 31
10. Else if month is 4, 6, 9, or 11 → Set max_day = 30
11. Else if month == 2 → If leap == True → Set max_day = 29
    → Else → Set max_day = 28
12. If day < 1 OR day > max_day → Print "Invalid Date" → Stop
13. Increment day by 1
14. If day > max_day → Set day = 1 → Increment month by 1
15. If month > 12 → Set month = 1 → Increment year by 1
16. Print "Next Date is: day/month/year"

17. Stop.


Start
Input date, month, year
valid_date = True
If(year <=0 or month >12 or month < 1)
Yes
No
valid_date = False
if month in (1,3,5,7,8,10 ,12)
Yes
No
max_date = 31
else if month in (4,6,9,11)
Yes
No
max_date = 30
If (year%4==0)
Yes
No
max_date = 29
max_date = 28
If date > max date or date < 1
Yes
No
valid_date = False


No
If valid_date
=True
Yes
Yes
If date <
max_date
No
date += 1
date = 1
Yes
If month==12
month = 1
year += 1
No
month +=1
Print date month year
Print Invalid Date
Stop